Pui In Mak

professor

基本信息

姓名： Pui In Mak

所属院校： 澳门大学

地区： China-Macau

学科领域

微电子科学与工程 | 集成电路设计与集成系统 | 通信工程 | 生物医学工程

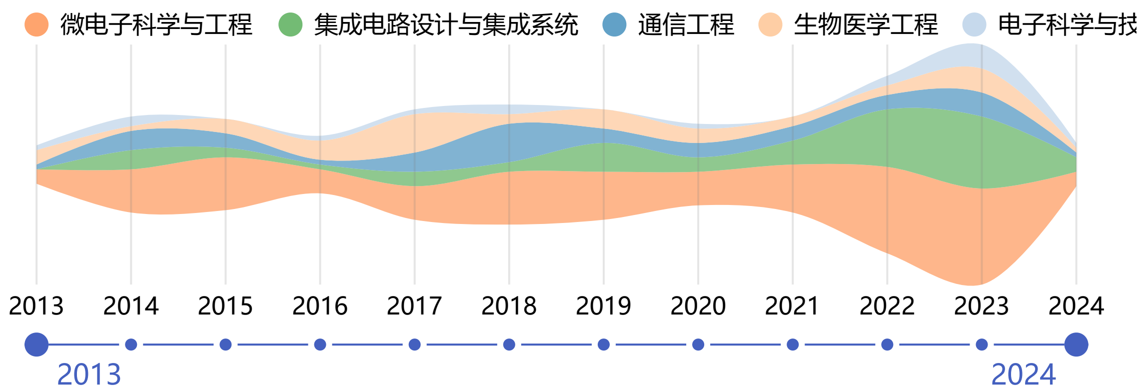
研究领域

射频集成电路 | 模拟与数模混合电路 | 低功耗电路与能量管理 | 微电子器件与制造 | 超大规模集成电路设计 | 生物医学传感与检测技术 | 模拟与射频IC设计

个人简介

Pui In Mak, also known as 麦沛然, is a distinguished Full Professor. His professional contact information includes the email address pimak and the phone number (+853) 8822-8794. He is based in room N21-3006d. Professor Mak has an extensive academic record, with 37 awards, 31 patents and technology transfers, 202 journal publications, 191 conference papers, and 8 books and book chapters to his credit. For a detailed curriculum vitae, visit http://www.fst.umac.mo/en/staff/fstpim.html.

近年发表的论文



A 28-nm Computing-in-Memory-Based Super-Resolution Accelerator Incorporating Macro-Level Pipeline and Texture/Algebraic Sparsity

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wu, Hao; Chen, Yong; Yuan, Yiyang; Yue, Jinshan; Fu, Xiangqu; Ren, Qirui; Luo, Qing; Mak, Pui-In; Wang, Xinghua; Zhang, Feng

被引用次数：1 2024

A Compact Sub-nW/kHz Relaxation Oscillator Using a Negative-Offset Comparator With Chopping and Piecewise Charge-Acceleration in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Yueduo; Zhu, Zihao; Bao, Rongxin; Lin, Jiahui; Yin, Jun; Li, Qiang; Mak, Pui-In; Yang, Shiheng

被引用次数：0 2024

A 5.6-dB Noise Figure, 63-86-GHz Receiver Using a Wideband Noise-Cancelling Low Noise Amplifier With Phase and Amplitude Compensation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Han, Changxuan; Deng, Zhixian; Shu, Yiyang; Yin, Jun; Mak, Pui-In; Luo, Xun

被引用次数：0 2024

Absolute quantification of nucleic acid on digital microfluidics platform based on superhydrophobic-superhydrophilic micropatterning

SENSORS AND ACTUATORS B-CHEMICAL

Meng, Li; Li, Mingzhong; Xu, Zhenyu; Lv, Aman; Jia, Yanwei; Chen, Meiwan; Mak, Pui-In; Martins, Rui P.; Law, Man-Kay

被引用次数：1 2024

A 93.4% Peak Efficiency CLOAD-Free Multi-Phase Switched-Capacitor DC-DC Converter Achieving a Fast DVS up to 222.5 mV/ns

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Feiyu; Fang, Qishen; Wu, Jiangchao; Jiang, Yang; Mak, Pui-In; Martins, Rui P.; Law, Man-Kay

被引用次数：0 2024

A 12-/13.56-MHz Crystal Oscillator With Binary-Search-Assisted Two-Step Injection Achieving 5.0-nJ Startup Energy and 45.8-μs Startup Time

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Haihua; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

被引用次数：2 2024

A 0.05-mm2 2.91-nJ/Decision Keyword-Spotting (KWS) Chip Featuring an Always-Retention 5T-SRAM in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tan, Fei; Yu, Wei-Han; Un, Ka-Fai; Martins, Rui P. P.; Mak, Pui-In

被引用次数：8 2024

A Security-Enhanced, Charge-Pump-Free, ISO14443-A-/ISO10373-6-Compliant RFID Tag With 16.2-μW Embedded RRAM and Reconfigurable Strong PUF

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ren, Qirui; Huo, Qiang; Chen, Zhisheng; Gao, Qi; Wang, Yiming; Yang, Yiming; Wu, Hao; Fu, Xiangqu; Xu, Xiaoxin; Luo, Qing; Gao, Jianfeng; Chen, Chengying; Zhao, Xiaojin; Lei, Dengyun; Wang, Xinghua; Zhang, Feng; Chen, Yong; Mak, Pui-In

被引用次数：1 2023

A syndromic diagnostic assay on a macrochannel-to-digital microfluidic platform for automatic identification of multiple respiratory pathogens

LAB ON A CHIP

Dong, Cheng; Li, Fei; Sun, Yun; Long, Dongling; Chen, Chunzhao; Li, Mengyan; Wei, Tao; Martins, Rui P.; Chen, Tianlan; Mak, Pui-In

被引用次数：2 2023

A 3.6-GHz Type-II Sampling PLL With a Differential Parallel-Series Double-Edge S-PD Scoring 43.1-fsRMSJitter,-258.7-dB FOM, and-75.17-dBc Reference Spur

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Huang, Yunbo; Chen, Yong; Zhao, Bo; Mak, Pui-In; Martins, Rui P.

被引用次数：4 2023

A 3.78-GHz Type-I Sampling PLL With a Fully Passive KPD-Doubled Master-Slave S-PD Measuring 39.6-fsRMS Jitter,-260.2-dB FOM, and-70.96-dBc Reference Spur

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Huang, Yunbo; Chen, Yong; Zhao, Bo; Mak, Pui-In; Martins, Rui P.

被引用次数：4 2023

On the DC-Settling Process of the Pierce Crystal Oscillator in Start-Up

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Zehao; Yang, Shiheng; Liu, Yueduo; Zhu, Zihao; Lin, Jiahui; Bao, Rongxin; Xu, Tailong; Yang, Zhizhan; Zhang, Mingkang; Liu, Jiaxin; Zhou, Xiong; Yin, Jun; Mak, Pui-In; Li, Qiang

被引用次数：0 2023

A Digital Readout Integrated Circuit Based on Pixel-Level ADC Incorporating On-Chip Image Algorithm Calibration for IRFPA

IEEE SENSORS JOURNAL

Zeng, Yan; Yang, Shiheng; Liu, Yueduo; Bao, Rongxin; Zhu, Zihao; Lin, Jiahui; Zhou, Xiong; Chen, Yong; Yin, Jun; Mak, Pui-In; Li, Qiang

被引用次数：1 2023

A 0.0043-mm2 0.085-μW/MHz Relaxation Oscillator Using Charge-Prestored Asymmetric Swings R-RC Network

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wei, Yuchen; Yang, Shiheng; Liu, Yueduo; Bao, Rongxin; Zhu, Zihao; Lin, Jiahui; Zhang, Zehao; Chen, Yong; Yin, Jun; Mak, Pui-In; Li, Qiang

被引用次数：1 2023

Nucleic acid analysis on electrowetting-based digital microfluidics

TRAC-TRENDS IN ANALYTICAL CHEMISTRY

Shen, Ren; Lv, A'man; Yi, Shuhong; Wang, Ping; Mak, Pui-In; Martins, Rui P.; Jia, Yanwei

被引用次数：6 2023

A 3.57-mW 2.88-GHz Multi-Phase Injection-Locked Ring-VCO With a 200-kHz 1/f3 Phase Noise Corner

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Fan, Chao; Zhao, Ya; Zhang, Yanlong; Yin, Jun; Geng, Li; Mak, Pui-In

被引用次数：5 2023

Modeling-Attack-Resistant Strong PUF Exploiting Stagewise Obfuscated Interconnections With Improved Reliability

IEEE INTERNET OF THINGS JOURNAL

Xu, Chongyao; Zhang, Litao; Law, Man-Kay; Zhao, Xiaojin; Mak, Pui-In; Martins, Rui P.

被引用次数：7 2023

---

pH Regulator on Digital Microfluidics with Pico-Dosing Technique

BIOSENSORS-BASEL

Li, Haoran; Peng, Tao; Zhong, Yunlong; Liu, Meiqing; Mak, Pui-In; Martins, Rui P.; Wang, Ping; Jia, Yanwei

被引用次数：0 2023

A 2.0-to-7.4-GHz 16-Phase Delay-Locked Loop With a Sub-0.6-ps Phase-Delay Error in 40-nm CMOS

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Yang, Jian; Pan, Quan; Yin, Jun; Mak, Pui-In

被引用次数：3 2023

A 1224 V-Input HV-LV-Separated Hybrid SC PoL Converter With 355 mWmm3 Power Density at 3 A Load Current and 15.2 mm3 Power Passives

IEEE TRANSACTIONS ON POWER ELECTRONICS

Zhang, Xiongjie; Ma, Qiaobo; Zhao, Anyang; Jiang, Yang; Law, Man-Kay; Jiang, Junmin; Takamiya, Makoto; Martins, Rui P.; Mak, Pui-In

被引用次数：4 2023

A Flexible Rooftop Photovoltaic-Inductive Wireless Power Transfer System for Low-Voltage DC Grid

IEEE ACCESS

Iam, Io-Wa; Ding, Zhaoyi; Huang, Zhicong; Lam, Chi-Seng; Martins, Rui P. P.; Mak, Pui-In

被引用次数：1 2023

A Subthreshold Operation Series-Parallel Charge Pump Incorporating Dynamic Source-Fed Oscillator for Wide-Input-Voltage Energy Harvesting Application

IEEE ACCESS

Kee, Yong Jack; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chong, Gabriel; Mekhilef, Saad; Lai, Nai Shyan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：2 2023

Design Trends and Perspectives of Digital Low Dropout Voltage Regulators for Low Voltage Mobile Applications: A Review

IEEE ACCESS

Lai, Li Fang; Ramiah, Harikrishnan; Tan, Yee-Chyan; Lai, Nai Shyan; Lim, Chee-Cheow; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：5 2023

A Fully-Integrated CMOS Dual-Band RF Energy Harvesting Front-End Employing Adaptive Frequency Selection

IEEE ACCESS

Lian, Wen Xun; Ramiah, Harikrishnan; Chong, Gabriel; Pakkirisami Churchill, Kishore Kumar; Lai, Nai Shyan; Mekhilef, Saad; Chen, Yong; Mak, Pui-In; Martins, Rui P. P.

被引用次数：10 2023

Design and Implementation of Hybrid DC-DC Converter: A Review

IEEE ACCESS

Chang, Teck Seong; Ramiah, Harikrishnan; Jiang, Yang; Lim, Chee Cheow; Lai, Nai Shyan; Mak, Pui-In; Martins, Rui P.

被引用次数：6 2023

High-Performance Multiband Ambient RF Energy Harvesting Front-End System for Sustainable IoT Applications-A Review

IEEE ACCESS

Lee, Yi Chen; Ramiah, Harikrishnan; Choo, Alexander; Churchill, Kishore Kumar Pakkirisami; Lai, Nai Shyan; Lim, Chee Cheow; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：17 2023

A 47-nW Voice Activity Detector (VAD) Featuring a Short-Time CNN Feature Extractor and an RNN-Based Classifier With a Non-Volatile CAP-ROM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Jinhai; Un, Ka-Fai; Yu, Wei-Han; Martins, Rui P.; Mak, Pui-In

被引用次数：4 2023

A 0.4-V 0.0294-mm2 Resistor-Based Temperature Sensor Achieving ±0.24 °C p2p Inaccuracy From-40°C to 125°C and 385 fJ • K2 Resolution FoM in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shi, Dan; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

被引用次数：3 2023

A Miniaturized 3-D-MRI Scanner Featuring an HV-SOI ASIC and Achieving a 10 x 8 x 8 mm3 Field of View

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fan, Shuhao; Zhou, Qi; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

被引用次数：8 2023

A 10.5 W, 93% Efficient Dual-Path Hybrid(DPH)-Based DC-DC Converter Incorporating a Continuous-Current-Input Switched-Capacitor Stage and Enhanced IL Reduction for 12 V/24 V Inputs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ma, Qiaobo; Zhang, Xiongjie; Zhao, Anyang; Li, Huihua; Jiang, Yang; Law, Man-Kay; Takamiya, Makoto; Martins, Rui P.; Mak, Pui-In

被引用次数：2 2023

Analysis and Design of a 15.2-to-18.2-GHz Inverse-Class-F VCO With a Balanced Dual-Core Topology Suppressing the Flicker Noise Upconversion

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Meng, Xi; Li, Haoran; Chen, Peng; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：2 2023

A Fully Integrated CMOS Tri-Band Ambient RF Energy Harvesting System for IoT Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yong, Jack Kee; Lian, Wen Xun; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chong, Gabriel; Lai, Nai Shyan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：8 2023

Floating-Domain Integrated GaN Driver Techniques for DC-DC Converters: A Review

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Mu, Xuchu; Zhao, Guangshu; Zhao, Anyang; Jiang, Yang; Law, Man-Kay; Takamiya, Makoto; Mak, Pui-In; Martins, Rui P.

被引用次数：4 2023

A 10.8-to-37.4 Gb/s Reference-Less FD-Less Single-Loop Quarter-Rate Bang-Bang Clock and Data Recovery Employing Deliberate-Current-Mismatch Wide-Frequency-Acquisition Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Lin; Chen, Yong; Yang, Chaowei; Zhao, Xiaoteng; Mak, Pui-In; Maloberti, Franco; Martins, Rui P.

被引用次数：1 2023

A 28-nm 368-fJ/Cycle, 0.43%/V Supply-Sensitivity, FLL-Based RC Oscillator Featuring Positive-TC-Only Resistors and ΔΣM-Based Trimming

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Huang, Yunbo; Chen, Yong; Yang, Kaiyuan; Crovetti, Paolo; Mak, Pui-In; Martins, Rui P.

被引用次数：2 2023

A High-PCE Range-Extension CMOS Rectifier Employing Advanced Topology Amalgamation Technique for Ambient RF Energy Harvesting

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choo, Alexander; Lee, Yi Chen; Ramiah, Harikrishnan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：10 2023

A 0.4-V 8400-m2 Voltage Reference in 65-nm CMOS Exploiting Well-Proximity Effect

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Che, Chengyu; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

被引用次数：2 2023

Universal Stability Criterion for Type-I Sampling Phase-Locked Loops

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Huang, Yunbo; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：3 2023

A 27-dBm, 0.92-GHz CMOS Power Amplifier With Mode Switching and a High-Q Compact Inductor (HQCI) Achieving a 30 Back-Off PAE

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Rawat, Arvind Singh; Rajendran, Jagadheswaran; Mariappan, Selvakumar; Kumar, Narendra; Mak, Pui-In; Martins, Rui P.

被引用次数：2 2023

An FPGA-Based Transformer Accelerator Using Output Block Stationary Dataflow for Object Recognition Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhao, Zhongyu; Cao, Rujian; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数：8 2023

A Reconfigurable CMOS Stack Rectifier With 22.8-dB Dynamic Range Achieving 47.91% Peak PCE for IoT/WSN Application

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Churchill, Kishore Kumar Pakkirisami; Ramiah, Harikrishnan; Choo, Alexander; Chong, Gabriel; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：4 2023

A High-Performance Dual-Topology CMOS Rectifier With 19.5-dB Power Dynamic Range for RF-Based Hybrid Energy Harvesting

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Choo, Alexander; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chen, Yong; Mekhilef, Saad; Mak, Pui-In; Martins, Rui P.

被引用次数：7 2023

A Reconfigurable Hybrid RF Front-End Rectifier for Dynamic PCE Enhancement of Ambient RF Energy Harvesting Systems

ELECTRONICS

Lian, Wen Xun; Yong, Jack Kee; Chong, Gabriel; Churchill, Kishore Kumar Pakkirisami; Ramiah, Harikrishnan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：8 2023

Sub-5-Minute Ultrafast PCR using Digital Microfluidics

BIOSENSORS & BIOELECTRONICS

Wan, Liang; Li, Mingzhong; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.; Jia, Yanwei

被引用次数：9 2023

A CMOS Hall Sensors Array With Integrated Readout Circuit Resilient to Local Magnetic Interference From Current-Carrying Traces

IEEE SENSORS JOURNAL

Zou, Hengchen; Lei, Ka-Meng; Martins, Rui P.; Mak, Pui-In

被引用次数：2 2023

An Ultra-Low-Voltage Single-Crystal Oscillator-Timer (XO-Timer) Delivering 16-MHz and 32.258-kHz Clocks for Sub-0.5-V Energy-Harvesting BLE Radios in 28-nm CMOS

IEEE OPEN JOURNAL OF CIRCUITS AND SYSTEMS

Lin, Liwen; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

被引用次数：0 2023

A 3.3-GHz Integer N-Type-II Sub-Sampling PLL Using a BFSK-Suppressed Push-Pull SS-PD and a Fast-Locking FLL Achieving-82.2-dBc REF Spur and-255-dB FOM

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yang, Zunsong; Chen, Yong; Yuan, Jia; Mak, Pui-In; Martins, Rui P.

被引用次数：19 2022

A Multimode CMOS Vision Sensor With On-Chip Motion Direction Detection and Simultaneous Energy Harvesting Capabilities (vol 22, pg 12808, 2022)

IEEE SENSORS JOURNAL

Wu, Jiangchao; Lu, Xin; Law, Man-Kay; Jiang, Yang; Liu, Liyuan; Mak, Pui-In; Martins, Rui P.

被引用次数：0 2022

A Multimode CMOS Vision Sensor With On-Chip Motion Direction Detection and Simultaneous Energy Harvesting Capabilities

IEEE SENSORS JOURNAL

Wu, Jiangchao; Lu, Xin; Law, Man-Kay; Jiang, Yang; Liu, Liyuan; Mak, Pui-In; Martins, Rui P.

被引用次数：3 2022

A 6-to-7.5-GHz 54-fsrms Jitter Type-II Reference-Sampling PLL Featuring a Gain-Boosting Phase Detector for In-Band Phase-Noise Reduction

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Xu, Tailong; Zhong, Shenke; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：5 2022

Accurate Performance Evaluation of Jitter-Power FOM for Multiplying Delay-Locked Loop

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Yueduo; Bao, Rongxin; Zhu, Zihao; Yang, Shiheng; Zhou, Xiong; Yin, Jun; Mak, Pui-In; Li, Qiang

被引用次数：1 2022

A Low-Jitter and Low-Reference-Spur 320 GHz Signal Source With an 80 GHz Integer-N Phase-Locked Loop Using a Quadrature XOR Technique

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Liang, Yuan; Boon, Chirn Chye; Qi, Gengzhen; Dziallas, Giannino; Kissinger, Dietmar; Ng, Herman Jalli; Mak, Pui-In; Wang, Yong

被引用次数：8 2022

A 266-μW Bluetooth Low-Energy (BLE) Receiver Featuring an N -Path Passive Balun-LNA and a Pipeline Down-Mixing BB-Extraction Scheme Achieving 77-dB SFDR and-3-dBm OOB-B-1dB

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shao, Haijun; Mak, Pui-In; Qi, Gengzhen; Martins, Rui P.

被引用次数：4 2022

A 1.7-3.6 GHz 20 MHz-Bandwidth Channel-Selection N-Path Passive-LNA Using a Switched-Capacitor-Transformer Network Achieving 23.5 dBm OB-IIP3 and 3.4-4.8 dB NF

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shao, Haijun; Qi, Gengzhen; Mak, Pui-In; Martins, Rui P.

被引用次数：13 2022

A 4T/Cell Amplifier-Chain-Based XOR PUF With Strong Machine Learning Attack Resilience

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Jieyun; Xu, Chongyao; Law, Man-Kay; Jiang, Yang; Zhao, Xiaojin; Mak, Pui-In; Martins, Rui P.

被引用次数：14 2022

Constant-Frequency and Noncommunication-Based Inductive Power Transfer Converter for Battery Charging

IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN POWER ELECTRONICS

Iam, Io-Wa; Hoi, Iok-U; Huang, Zhicong; Gong, Cheng; Lam, Chi-Seng; Mak, Pui-In; Da Silva Martins, Rui Paulo

被引用次数：14 2022

Ratiometric fluorescence analysis for miR-141 detection with hairpin DNA-templated silver nanoclusters

JOURNAL OF MATERIALS CHEMISTRY C

Liu, Meiqing; Shen, Ren; Li, Haoran; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：11 2022

Low Voltage Switched-Capacitive-Based Reconfigurable Charge Pumps for Energy Harvesting Systems: An Overview

IEEE ACCESS

Ho, Tian Siang; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chen, Yong; Lim, Chee Cheow; Lai, Nai Shyan; Mak, Pui-In; Martins, Rui P. P.

被引用次数：12 2022

Evaluation and Perspective of Analog Low-Dropout Voltage Regulators: A Review

IEEE ACCESS

Chyan, Tan Yee; Ramiah, Harikrishnan; Hatta, S. F. Wan Muhamad; Lai, Nai Shyan; Lim, Chee-Cheow; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：12 2022

Fully-Integrated Timers for Ultra-Low-Power Internet-of-Things Nodes-Fundamentals and Design Techniques

IEEE ACCESS

Loo, Mikki How-Wen; Ramiah, Harikrishnan; Lei, Ka-Meng; Lim, Chee Cheow; Lai, Nai Shyan; Mak, Pui-In; Martins, Rui P.

被引用次数：6 2022

A 108-nW 0.8-mm2 Analog Voice Activity Detector Featuring a Time-Domain CNN With Sparsity-Aware Computation and Sparsified Quantization in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Feifei; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数：8 2022

Arithmetic Progression Switched-Capacitor DC-DC Converter Topology With Soft VCR Transitions and Quasi-Symmetric Two-Phase Charge Delivery

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Yang; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

被引用次数：6 2022

A Sub-0.25-pJ/bit 47.6-to-58.8-Gb/s Reference-Less FD-Less Single-Loop PAM-4 Bang-Bang CDR With a Deliberate-Current-Mismatch Frequency Acquisition Technique in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Xiaoteng; Chen, Yong; Wang, Lin; Mak, Pui-In; Maloberti, Franco; Martins, Rui P.

被引用次数：10 2022

A 0.0285-mm2 0.68-pJ/bit Single-Loop Full-Rate Bang-Bang CDR Without Reference and Separate FD Pulling Off an 8.2-Gb/s/μs Acquisition Speed of the PAM-4 Input in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Xiaoteng; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：13 2022

Miniaturization of a Nuclear Magnetic Resonance System: Architecture and Design Considerations of Transceiver Integrated Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Fan, Shuhao; Zhou, Qi; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

被引用次数：7 2022

Mismatch Analysis of DTCs With an Improved BIST-TDC in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Peng; Yin, Jun; Zhang, Feifei; Mak, Pui-In; Martins, Rui P.; Staszewski, Robert Bogdan

被引用次数：6 2022

A Low-Power Multiband Blocker-Tolerant Receiver With a Steep Filtering Slope Using an N-Path LNA With Feedforward OB Blocker Cancellation and Filtering-by-Aliasing Baseband Amplifiers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shao, Haijun; Qi, Gengzhen; Mak, Pui-In; Martins, Rui P.

被引用次数：16 2022

A Millimeter-Wave CMOS VCO Featuring a Mode-Ambiguity-Aware Multi-Resonant-RLCM Tank

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Guo, Hao; Chen, Yong; Yang, Chaowei; Mak, Pui-In; Martins, Rui P.

被引用次数：25 2022

A 529-μW Fractional-N All-Digital PLL Using TDC Gain Auto-Calibration and an Inverse-Class-F DCO in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Peng; Meng, Xi; Yin, Jun; Mak, Pui-In; Martins, Rui P.; Staszewski, Robert Bogdan

被引用次数：16 2022

One-shot high-resolution melting curve analysis for KRAS point-mutation discrimination on a digital microfluidics platform

LAB ON A CHIP

Li, Mingzhong; Wan, Liang; Law, Man-Kay; Meng, Li; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：14 2022

A 0.1-V VIN Subthreshold 3-Stage Dual-Branch Charge Pump With 43.4% Peak Power Conversion Efficiency Using Advanced Dynamic Gate-Bias

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yong, Jack Kee; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chong, Gabriel; Mekhilef, Saad; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：16 2022

High-Performance Harmonic-Rich Single-Core VCO With Multi-LC Tank: A Tutorial

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：36 2022

Miniaturized Energy Harvesting Systems Using Switched-Capacitor DC-DC Converters

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Law, Man-Kay; Jiang, Yang; Mak, Pui-In; Martins, Rui P.

被引用次数：8 2022

A-20-dBm Sensitivity RF Energy-Harvesting Rectifier Front End Using a Transformer IMN

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lian, Wen Xun; Ramiah, Harikrishnan; Chong, Gabriel; Churchill, Kishore Kumar Pakkirisami; Lai, Nai Shyan; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：13 2022

A Reconfigurable CMOS Rectifier With 14-dB Power Dynamic Range Achieving >36-dB/mm2 FoM for RF-Based Hybrid Energy Harvesting

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Choo, Alexander; Ramiah, Harikrishnan; Churchill, Kishore Kumar Pakkirisami; Chen, Yong; Mekhilef, Saad; Mak, Pui-In; Martins, Rui P.

被引用次数：19 2022

A Fully-Integrated Ambient RF Energy Harvesting System with 423-μW Output Power

SENSORS

Churchill, Kishore Kumar Pakkirisami; Ramiah, Harikrishnan; Chong, Gabriel; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：16 2022

A 3.36-GHz Locking-Tuned Type-I Sampling PLL With-78.6-dBc Reference Spur Merging Single-Path Reference-Feedthrough-Suppression and Narrow-Pulse-Shielding Techniques

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Huang, Yunbo; Chen, Yong; Jiao, Hailong; Mak, Pui-In; Martins, Rui P.

被引用次数：20 2021

A 600-μm2 Ring-VCO-Based Hybrid PLL Using a 30-μW Charge-Sharing Integrator in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yang, Shiheng; Yin, Jun; Xu, Tailong; Yi, Taimo; Mak, Pui-In; Li, Qiang; Martins, Rui P.

被引用次数：6 2021

Cancer drug screening with an on-chip multi-drug dispenser in digital microfluidics

LAB ON A CHIP

Zhai, Jiao; Li, Caiwei; Li, Haoran; Yi, Shuhong; Yang, Ning; Miao, Kai; Deng, Chuxia; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：21 2021

Temperature Tolerance Electric Cell-Substrate Impedance Sensing for Joint Assessment of Cell Viability and Vitality

ACS SENSORS

Yang, Ning; Hui, Wenhao; Dong, Sizhe; Zhang, Xuming; Shao, Liyang; Jia, Yanwei; Mak, Pui-In; Martins, Rui Paulo da Silva

被引用次数：3 2021

Wideband Variable-Gain Amplifiers Based on a Pseudo-Current-Steering Gain-Tuning Technique

IEEE ACCESS

Kong, Lingshan; Chen, Yong; Yu, Haohong; Boon, Chirn Chye; Mak, Pui-In; Martins, Rui P.

2021

A 0.35-V 5,200-μm2 2.1-MHz Temperature-Resilient Relaxation Oscillator With 667 fJ/Cycle Energy Efficiency Using an Asymmetric Swing-Boosted RC Network and a Dual-Path Comparator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

2021

A 0.003-mm2 440fsRMS-Jitter and-64dBc-Reference-Spur Ring-VCO-Based Type-I PLL Using a Current-Reuse Sampling Phase Detector in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yang, Zunsong; Chen, Yong; Mak, Pui-In; Martins, Rui P.

2021

A 0.14-to-0.29-pJ/bit 14-GBaud/s Trimodal (NRZ/PAM-4/PAM-8) Half-Rate Bang-Bang Clock and Data Recovery (BBCDR) Circuit in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhao, Xiaoteng; Chen, Yong; Mak, Pui-In; Martins, Rui P.

2021

A 1.7-to-2.7GHz 35-38% PAE Multiband CMOS Power Amplifier Employing a Digitally-Assisted Analog Pre-Distorter (DAAPD) Reconfigurable Linearization Technique

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mariappan, Selvakumar; Rajendran, Jagadheswaran; Chen, Yong; Mak, Pui-In; Martins, Rui P.

2021

An FPGA-Based Energy-Efficient Reconfigurable Convolutional Neural Network Accelerator for Object Recognition Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Li, Jixuan; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

2021

A Wide-PCE-Dynamic-Range CMOS Cross-Coupled Differential-Drive Rectifier for Ambient RF Energy Harvesting

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chong, Gabriel; Ramiah, Harikrishnan; Yin, Jun; Rajendran, Jagadheswaran; Mak, Pui-In; Martins, Rui P.

2021

An 800 MHz-to-3.3 GHz 20-MHz Channel Bandwidth WPD CMOS Power Amplifier For Multiband Uplink Radio Transceivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mariappan, Selvakumar; Rajendran, Jagadheswaran; Ramiah, Harikrishnan; Mak, Pui-In; Yin, Jun; Martins, Rui P.

被引用次数：4 2021

Startup Time and Energy-Reduction Techniques for Crystal Oscillators in the IoT Era

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.

被引用次数：12 2021

A Fully Integrated 10-V Pulse Driver Using Multiband Pulse-Frequency Modulation in 65-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wu, Jiangchao; Leong, Hou-Man; Jiang, Yang; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

被引用次数：0 2021

SARS-CoV-2 RNA Detection with Duplex-Specific Nuclease Signal Amplification

MICROMACHINES

Liu, Meiqing; Li, Haoran; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：9 2021

Cost-Effective Compensation Design for Output Customization and Efficiency Optimization in Series/Series-Parallel Inductive Power Transfer Converter

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Huang, Zhicong; Fang, Zhijian; Lam, Chi-Seng; Mak, Pui-In; Martins, Rui P.

被引用次数：10 2020

A digital microfluidic system with 3D microstructures for single-cell culture

MICROSYSTEMS & NANOENGINEERING

Zhai, Jiao; Li, Haoran; Wong, Ada Hang-Heng; Dong, Cheng; Yi, Shuhong; Jia, Yanwei; Mak, Pui-In; Deng, Chu-Xia; Martins, Rui P.

被引用次数：47 2020

A 4-μm Diameter SPAD Using Less-Doped N-Well Guard Ring in Baseline 65-nm CMOS

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lu, Xin; Law, Man-Kay; Jiang, Yang; Zhao, Xiaojin; Mak, Pui-In; Martins, Rui P.

被引用次数：14 2020

A Single-Stage Inductive-Power-Transfer Converter for Constant-Power and Maximum-Efficiency Battery Charging

IEEE TRANSACTIONS ON POWER ELECTRONICS

Huang, Zhicong; Lam, Chi-Seng; Mak, Pui-In; Martins, Rui Paulo da Silva; Wong, Siu-Chung; Tse, Chi K.

被引用次数：63 2020

Piezoelectric Energy-Harvesting Interface Using Split-Phase Flipping-Capacitor Rectifier With Capacitor Reuse for Input Power Adaptation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Zhiyuan; Law, Man-Kay; Mak, Pui-In; Zeng, Xiaoyang; Martins, Rui P.

被引用次数：24 2020

A 10.6-mW 26.4-GHz Dual-Loop Type-II Phase-Locked Loop Using Dynamic Frequency Detector and Phase Detector

IEEE ACCESS

Yang, Zunsong; Chen, Yong; Yang, Shiheng; Mak, Pui-In; Martins, Rui P.

被引用次数：20 2020

Portable NMR with Parallelism

ANALYTICAL CHEMISTRY

Lei, Ka-Meng; Ha, Dongwan; Song, Yi-Qiao; Westervelt, Robert M.; Martins, Rui; Mak, Pui-In; Ham, Donhee

被引用次数：31 2020

A Multiband FDD SAW-Less Transmitter for 5G-NR Featuring a BW-Extended N-Path Filter-Modulator, a Switched-BB Input, and a Wideband TIA-Based PA Driver

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qi, Gengzhen; Shao, Haijun; Mak, Pui-In; Yin, Jun; Martins, Rui P.

被引用次数：7 2020

A Single-Pin Antenna Interface RF Front End Using a Single-MOS DCO-PA and a Push-Pull LNA

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xu, Kai; Yin, Jun; Mak, Pui-In; Staszewski, Robert Bogdan; Martins, Rui P.

被引用次数：14 2020

A 0.096-mm2 1-20-GHz Triple-Path Noise-Canceling Common-Gate Common-Source LNA With Dual Complementary pMOS-nMOS Configuration

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Yu, Haohong; Chen, Yong; Boon, Chirn Chye; Mak, Pui-In; Martins, Rui P.

被引用次数：69 2020

A 3.3-mW 25.2-to-29.4-GHz Current-Reuse VCO Using a Single-Turn Multi-Tap Inductor and Differential-Only Switched-Capacitor Arrays With a 187.6-dBc/Hz FOM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Huang, Yunbo; Chen, Yong; Guo, Hao; Mak, Pui-In; Martins, Rui P.

被引用次数：43 2020

Design of a 4.2-to-5.1 GHz Ultralow-Power Complementary Class-B/C Hybrid-Mode VCO in 65-nm CMOS Fully Supported by EDA Tools

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Martins, Ricardo; Lourenco, Nuno; Horta, Nuno; Zhong, Shenke; Yin, Jun; Mak, Pui In; Martins, Rui P.

被引用次数：16 2020

Turning on/off satellite droplet ejection for flexible sample delivery on digital microfluidics

LAB ON A CHIP

Li, Haoran; Shen, Ren; Dong, Cheng; Chen, Tianlan; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：19 2020

Clip-to-release on amplification (CRoA): a novel DNA amplification enhancer on and off microfluidics

LAB ON A CHIP

Shen, Ren; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：6 2020

Design Considerations of the Interpolative Digital Transmitter for Quantization Noise and Replicas Rejection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Un, Ka-Fai; Zhang, Feifei; Mak, Pui-In; Martins, Rui P.; Zhu, Anding; Staszewski, Robert Bogdan

被引用次数：5 2020

Cell-based drug screening on microfluidics

TRAC-TRENDS IN ANALYTICAL CHEMISTRY

Zhai, Jiao; Yi, Shuhong; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：39 2019

Algebraic Series-Parallel-Based Switched-Capacitor DC-DC Boost Converter With Wide Input Voltage Range and Enhanced Power Density

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Yang; Law, Man-Kay; Chen, Zhiyuan; Mak, Pui-In; Martins, Rui P.

被引用次数：26 2019

Hydrodynamic-flow-enhanced rapid mixer for isothermal DNA hybridization kinetics analysis on digital microfluidics platform

SENSORS AND ACTUATORS B-CHEMICAL

Li, Mingzhong; Dong, Cheng; Law, Man-Kay; Jia, Yanwei; Mak, Pui-In; Martins, Rui P.

被引用次数：13 2019

A 0.2-V Energy-Harvesting BLE Transmitter With a Micropower Manager Achieving 25% System Efficiency at 0-dBm Output and 5.2-nW Sleep Power in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Shiheng; Yin, Jun; Yi, Haidong; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数：45 2019

A 0.0056-mm2-249-dB-FoM All-Digital MDLL Using a Block-Sharing Offset-Free Frequency-Tracking Loop and Dual Multiplexed-Ring VCOs

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Shiheng; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：28 2019

A 40-Gb/s PAM-4 Transmitter Using a 0.16-pJ/bit SST-CML-Hybrid (SCH) Output Driver and a Hybrid-Path 3-Tap FFE Scheme in 28-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Fan, Chao; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数：12 2019

A 0.0071-mm2 10.8pspp-Jitter 4 to 10-Gb/s 5-Tap Current-Mode Transmitter Using a Hybrid Delay Line for Sub-1-UI Fractional De-Emphasis

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Yong; Mak, Pui-In; Yang, Zunsong; Boon, Chirn Chye; Martins, Rui P.

被引用次数：9 2019

A 0.12-mm2 1.2-to-2.4-mW 1.3-to-2.65-GHz Fractional-N Bang-Bang Digital PLL With 8-μs Settling Time for Multi-ISM-Band ULP Radios

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Un, Ka-Fai; Qi, Gengzhen; Yin, Jun; Yang, Shiheng; Yu, Shupeng; Ieong, Chio-In; Mak, Pui-In; Martins, Rui P.

被引用次数：11 2019

A 0.0018-mm2 153% Locking-Range CML-Based Divider-by-2 With Tunable Self-Resonant Frequency Using an Auxiliary Negative-gm Cell

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhao, Xiaoteng; Chen, Yong; Mak, Pui-In; Martins, Rui P.

被引用次数：14 2019

CMOS Cross-Coupled Differential-Drive Rectifier in Subthreshold Operation for Ambient RF Energy Harvesting-Model and Analysis

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chong, Gabriel; Ramiah, Harikrishnan; Yin, Jun; Rajendran, Jagadheswaran; Wong, Wei Ru; Mak, Pui-In; Martins, Rui P.

被引用次数：24 2019

Fully Integrated High Voltage Pulse Driver Using Switched-Capacitor Voltage Multiplier and Synchronous Charge Compensation in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Wu, Jiangchao; Lei, Ka-Chon; Leong, Hou-Man; Jiang, Yang; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

被引用次数：10 2019

A Comparative Study of 8-Phase Feedforward-Coupling Ring VCOs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sun, Iat-Fai; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：14 2019

A 0.5-V 0.4-to-1.6-GHz 8-Phase Bootstrap Ring-VCO Using Inherent Non-Overlapping Clocks Achieving a 162.2-dBc/Hz FoM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jiang, Tongquan; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：12 2019

A 5.1-to-7.3 mW, 2.4-to-5 GHz Class-C Mode-Switching Single-Ended-Complementary VCO Achieving > 190 dBc/Hz FoM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lim, Chee Cheow; Ramiah, Harikrishnan; Yin, Jun; Kumar, Narendra; Mak, Pui-In; Martins, Rui P.

 2019

A 0.044-mm2 0.5-to-7-GHz Resistor-Plus-Source-Follower-Feedback Noise-Cancelling LNA Achieving a Flat NF of 3.3±0.45 dB

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yu, Haohong; Chen, Yong; Boon, Chirn Chye; Li, Chenyang; Mak, Pui-In; Martins, Rui P.

 2019

Micro- and nanofabrication NMR technologies for point-of-care medical applications - A review

MICROELECTRONIC ENGINEERING

Dupre, Antoine; Lei, Ka-Meng; Mak, Pui-In; Martins, Rui P.; Peng, Weng Kung

 2019

Guest Editorial Special Issue on Magnetic Sensing Systems for Biomedical Application

IEEE SENSORS JOURNAL

Heidari, Hadi; Mak, Pui-In; Anders, Jens; Hall, Drew

 2019

Algorithmic Voltage-Feed-In Topology for Fully Integrated Fine-Grained Rational Buck-Boost Switched-Capacitor DC-DC Converters

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Yang; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

 2018

An Inverse-Class-F CMOS Oscillator With Intrinsic-High-Q First Harmonic and Second Harmonic Resonances

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lim, Chee Cheow; Ramiah, Harikrishnan; Yin, Jun; Mak, Pui-In; Martins, Rui P.

 2018

Low-Phase-Noise Wideband Mode-Switching Quad-Core-Coupled mm-wave VCO Using a Single-Center-Tapped Switched Inductor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Peng, Yatao; Yin, Jun; Mak, Pui-In; Martins, Rui P.

 2018

A Regulation-Free Sub-0.5-V 16-/24-MHz Crystal Oscillator With 14.2-nJ Startup Energy and 31.8-μW Steady-State Power

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

 2018

A 0.18-V 382-μW Bluetooth Low-Energy Receiver Front-End With 1.33-nW Sleep Power for Energy-Harvesting Applications in 28-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yi, Haidong; Yu, Wei-Han; Mak, Pui-In; Yin, Jun; Martins, Rui P.

被引用次数：56 2018

A SAW-Less Tunable RF Front End for FDD and IBFD Combining an Electrical-Balance Duplexer and a Switched-LC N-Path LNA

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qi, Gengzhen; van Liempd, Barend; Mak, Pui-In; Martins, Rui P.; Craninckx, Jan

被引用次数：76 2018

Introduction to the January Special Issue on the 2017 IEEE International Solid-State Circuits Conference

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bowman, Keith A.; Khellah, Muhammad M.; Kono, Takashi; Shor, Joseph; Mak, Pui-In

被引用次数：0 2018

A Wideband Inductorless dB-L near Automatic Gain Control Amplifier Using a Single-Branch Negative Exponential Generator for Wireline Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kong, Lingshan; Chen, Yong; Boon, Chirn Chye; Mak, Pui-In; Martins, Rui P.

被引用次数：29 2018

A Hardware-Efficient Feedback Polynomial Topology for DPD Linearization of Power Amplifiers: Theory and FPGA Validation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cheang, Chak-Fong; Mak, Pui-In; Martins, Rui P.

被引用次数：14 2018

A 36-Gb/s 1.3-mW/Gb/s Duobinary-Signal Transmitter Exploiting Power-Efficient Cross-Quadrature Clocking Multiplexers With Maximized Timing Margin

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Yong; Mak, Pui-In; Boon, Chirn Chye; Martins, Rui P.

被引用次数：27 2018

A 0.7-2.5 GHz, 61% EIRP System Efficiency, Four-Element MIMO TX System Exploiting Integrated Power-Relaxed Power Amplifiers and an Analog Spatial De-Interleaver

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Wei-Han; Un, Ka-Fai; Mak, Pui-In; Martins, Rui P.

被引用次数：3 2018

A Coin-Battery-Powered LDO-Free 2.4-GHz Bluetooth Low-Energy Transmitter With 34.7% Peak System Efficiency

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Peng, Xingqiang; Yin, Jun; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数：10 2018

A 0.032-mm2 0.15-V Three-Stage Charge-Pump Scheme Using a Differential Bootstrapped Ring-VCO for Energy-Harvesting Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Yi, Haidong; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：57 2018

Overview of Recent Development on Wireless Sensing Circuits and Systems for Healthcare and Biomedical Applications

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Li, Changzhi; Un, Ka-Fai; Mak, Pui-in; Chen, Ying; Munoz-Ferreras, Jose-Maria; Yang, Zhi; Gomez-Garcia, Roberto

被引用次数：33 2018

Guest Editorial Wireless Sensing Circuits and Systems for Healthcare and Biomedical Applications

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Li, Changzhi; Mak, Pui-In; Gomez-Garcia, Roberto; Chen, Ying

被引用次数：5 2018

Nano-Watt Class Energy-Efficient Capacitive Sensor Interface With On-Chip Temperature Drift Compensation

IEEE SENSORS JOURNAL

Zhang, Tan-Tan; Law, Man-Kay; Mak, Pui-In; Vai, Mang-I; Martins, Rui P.

被引用次数：17 2018

Fully Integrated Inductor-Less Flipping-Capacitor Rectifier for Piezoelectric Energy Harvesting

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Zhiyuan; Law, Man-Kay; Mak, Pui-In; Ki, Wing-Hung; Martins, Rui P.

被引用次数：83 2017

A 0.038-mm2 SAW-Less Multiband Transceiver Using an N-Path SC Gain Loop

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qi, Gengzhen; Mak, Pui-In; Martins, Rui P.

被引用次数：13 2017

A 2.4-GHz ZigBee Transmitter Using a Function-Reuse Class-F DCO-PA and an ADPLL Achieving 22.6% (14.5%) System Efficiency at 6-dBm (0-dBm) Pout

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Peng, Xingqiang; Yin, Jun; Mak, Pui-In; Yu, Wei-Han; Martins, Rui P.

被引用次数：24 2017

A Handheld High-Sensitivity Micro-NMR CMOS Platform With B-Field Stabilization for Multi-Type Biological/Chemical Assays

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Heidari, Hadi; Mak, Pui-In; Law, Man-Kay; Maloberti, Franco; Martins, Rui P.

被引用次数：42 2017

An Area-Efficient and Tunable Bandwidth-Extension Technique for a Wideband CMOS Amplifier Handling 50+Gb/s Signaling

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Chen, Yong; Mak, Pui-In; Yu, Haohong; Boon, Chirn Chye; Martins, Rui P.

被引用次数：31 2017

A 0.35-V 520-μW 2.4-GHz Current-Bleeding Mixer With Inductive-Gate and Forward-Body Bias, Achieving > 13-dB Conversion Gain and > 55-dB Port-to-Port Isolation

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Tan, Gim Heng; Ramiah, Harikrishnan; Mak, Pui-In; Martins, Rui P.

被引用次数：16 2017

A High-Voltage-Enabled Class-D Polar PA Using Interactive AM-AM Modulation, Dynamic Matching, and Power-Gating for Average PAE Enhancement

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Wei-Han; Peng, Xingqiang; Mak, Pui-In; Martins, Rui P.

被引用次数：5 2017

A 73.9%-Efficiency CMOS Rectifier Using a Lower DC Feeding (LDCF) Self-Body-Biasing Technique for Far-Field RF Energy-Harvesting Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Moghaddam, Amin Khalili; Chuah, Joon Huang; Ramiah, Harikrishnan; Ahmadian, Jalil; Mak, Pui-In; Martins, Rui P.

被引用次数：90 2017

A 3D microblade structure for precise and parallel droplet splitting on digital microfluidic chips

LAB ON A CHIP

Dong, Cheng; Jia, Yanwei; Gao, Jie; Chen, Tianlan; Mak, Pui-In; Vai, Mang-I; Martins, Rui P.

被引用次数：35 2017

A 27-Gb/s Time-Interleaved Duobinary Transmitter Achieving 1.44-mW/Gb/s FOM in 65-nm CMOS

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Chen, Yong; Mak, Pui-In; Boon, Chirn Chye; Martins, Rui P.

被引用次数：16 2017

A 0.45 V 147-375 nW ECG Compression Processor With Wavelet Shrinkage and Adaptive Temporal Decimation Architectures

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ieong, Chio-In; Li, Mingzhong; Law, Man-Kay; Mak, Pui-In; Vai, Mang I.; Martins, Rui P.

被引用次数：26 2017

A Single-Chip Solar Energy Harvesting IC Using Integrated Photodiodes for Biomedical Implant Applications

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Chen, Zhiyuan; Law, Man-Kay; Mak, Pui-In; Martins, Rui P.

被引用次数：82 2017

An Integrated Circuit for Simultaneous Extracellular Electrophysiology Recording and Optogenetic Neural Manipulation

IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING

Chen, Chang Hao; McCullagh, Elizabeth A.; Pun, Sio Hang; Mak, Peng Un; Vai, Mang I.; Mak, Pui In; Klug, Achim; Lei, Tim C.

被引用次数：40 2017

A digital microfluidic system for loop-mediated isothermal amplification and sequence specific pathogen detection

SCIENTIFIC REPORTS

Wan, Liang; Chen, Tianlan; Gao, Jie; Dong, Cheng; Wong, Ada Hang-Heng; Jia, Yanwei; Mak, Pui-In; Deng, Chu-Xia; Martins, Rui P.

被引用次数：64 2017

Drug screening of cancer cell lines and human primary tumors using droplet microfluidics

SCIENTIFIC REPORTS

Wong, Ada Hang-Heng; Li, Haoran; Jia, Yanwei; Mak, Pui-In; da Silva Martins, Rui Paulo; Liu, Yan; Vong, Chi Man; Wong, Hang Cheong; Wong, Pak Kin; Wang, Haitao; Sun, Heng; Deng, Chu-Xia

被引用次数：70 2017

A Time-Interleaved Ring-VCO with Reduced 1/f3 Phase Noise Corner, Extended Tuning Range and Inherent Divided Output

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yin, Jun; Mak, Pui-In; Maloberti, Franco; Martins, Rui P.

被引用次数：28 2016

A μNMR CMOS Transceiver Using a Butterfly-Coil Input for Integration With a Digital Microfluidic Device Inside a Portable Magnet

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

被引用次数：22 2016

A 2-μm InGaP/GaAs Class-J Power Amplifier for Multi-Band LTE Achieving 35.8-dB Gain, 40.5% to 55.8% PAE and 28-dBm Linear Output Power

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Jagadheswaran, U. R.; Ramiah, Harikrishnan; Mak, Pui-In; Martins, Rui P.

被引用次数：26 2016

Sub-7-second genotyping of single-nucleotide polymorphism by high-resolution melting curve analysis on a thermal digital microfluidic device

LAB ON A CHIP

Chen, Tianlan; Jia, Yanwei; Dong, Cheng; Gao, Jie; Mak, Pui-In; Martins, Rui P.

被引用次数：31 2016

CMOS biosensors for in vitro diagnosis - transducing mechanisms and applications

LAB ON A CHIP

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

2016

Design of a Collapse-Mode CMUT With an Embossed Membrane for Improving Output Pressure

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Yu, Yuanyu; Pun, Sio Hang; Mak, Peng Un; Cheng, Ching-Hsiang; Wang, Jiujiang; Mak, Pui-In; Vai, Mang I.

2016

Improved Analytical Modeling of Membrane Large Deflection With Lateral Force for the Underwater CMUT Based on Von Karman Equations

IEEE SENSORS JOURNAL

Wang, Jiujiang; Pun, Sio Hang; Mak, Peng Un; Cheng, Ching-Hsiang; Yu, Yuanyu; Mak, Pui-In; Vai, Mang I.

2016

A 1.1 μW CMOS Smart Temperature Sensor With an Inaccuracy of ±0.2 °C (3σ) for Clinical Temperature Monitoring

IEEE SENSORS JOURNAL

Law, Man-Kay; Lu, Sanfeng; Wu, Tao; Bermak, Amine; Mak, Pui-In; Martins, Rui P.

2016

A Highly-Scalable Analog Equalizer Using a Tunable and Current-Reusable Active Inductor for 10-Gb/sI/O Links

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chen, Yong; Mak, Pui-In; Wang, Yan

2015

A palm-size μNMR relaxometer using a digital microfluidic device and a semiconductor transceiver for chemical/biological diagnosis

ANALYST

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

2015

Adhesion promoter for a multi-dielectric-layer on a digital microfluidic chip

RSC ADVANCES

Gao, Jie; Chen, Tianlan; Dong, Cheng; Jia, Yanwei; Mak, Pui-In; Vai, Mang-I.; Martin, Rui P.

2015

Nested-Current-Mirror Rail-to-Rail-Output Single-Stage Amplifier With Enhancements of DC Gain, GBW and Slew Rate

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Zushu; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.; Maloberti, Franco

2015

A 0.02 mm2 59.2 dB SFDR 4th-Order SC LPF With 0.5-to-10 MHz Bandwidth Scalability Exploiting a Recycling SC-Buffer Biquad (vol 50, pg 1988, 2015)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mohan, P. V. Ananda; Zhao, Yaohua; Mak, Pui-In; Martins, Rui P.; Maloberti, Franco

被引用次数：0 2015

A 0.02 mm2 59.2 dB SFDR 4th-Order SC LPF With 0.5-to-10 MHz Bandwidth Scalability Exploiting a Recycling SC-Buffer Biquad

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhao, Yaohua; Mak, Pui-In; Martins, Rui P.; Maloberti, Franco

被引用次数：6 2015

A Sub-GHz Wireless Transmitter Utilizing a Multi-Class-Linearized PA and Time-Domain Wideband-Auto I/Q-LOFT Calibration for IEEE 802.11af WLAN

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Un, Ka-Fai; Yu, Wei-Han; Cheang, Chak-Fong; Qi, Gengzhen; Mak, Pui-In; Martins, Rui P.

被引用次数：9 2015

A Combinatorial Impairment-Compensation Digital Predistorter for a Sub-GHz IEEE 802.11af-WLAN CMOS Transmitter Covering a 10x-Wide RF Bandwidth

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cheang, Chak-Fong; Un, Ka-Fai; Yu, Wei-Han; Mak, Pui-In; Martins, Rui P.

被引用次数：10 2015

Wideband Receivers: Design Challenges, Tradeoffs and State-of-the-Art

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Lin, Fujian; Mak, Pui-In; Martins, Rui P.

被引用次数：36 2015

A 0.07 mm2 2.2 mW 10 GHz Current-Reuse Class-B/C Hybrid VCO Achieving 196-dBc/Hz FoMA

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Amin, Md. Tawfiq; Yin, Jun; Mak, Pui-In; Martins, Rui P.

被引用次数：22 2015

Improving the Linearity and Power Efficiency of Active Switched-Capacitor Filters in a Compact Die Area

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhao, Yaohua; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

被引用次数：1 2015

Energy Optimized Subthreshold VLSI Logic Family With Unbalanced Pull-Up/Down Network and Inverse Narrow-Width Techniques

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Li, Ming-Zhong; Ieong, Chio-In; Law, Man-Kay; Mak, Pui-In; Vai, Mang-I; Pun, Sio-Hang; Martins, Rui P.

被引用次数：18 2015

On the droplet velocity and electrode lifetime of digital microfluidics: voltage actuation techniques and comparison

MICROFLUIDICS AND NANOFLUIDICS

Dong, Cheng; Chen, Tianlan; Gao, Jie; Jia, Yanwei; Mak, Pui-In; Vai, Mang-I; Martins, Rui P.

被引用次数：36 2015

A 0.002-mm2 6.4-mW 10-Gb/s Full-Rate Direct DFE Receiver With 59.6% Horizontal Eye Opening Under 23.3-dB Channel Loss at Nyquist Frequency

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Chen, Yong; Mak, Pui-In; Zhang, Li; Wang, Yan

被引用次数：18 2014

NMR-DMF: a modular nuclear magnetic resonance-digital microfluidics system for biological assays

ANALYST

Lei, Ka-Meng; Mak, Pui-In; Law, Man-Kay; Martins, Rui P.

被引用次数：21 2014

A Sub-GHz Multi-ISM-Band ZigBee Receiver Using Function-Reuse and Gain-Boosted N-Path Techniques for IoT Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

被引用次数：57 2014

An RF-to-BB-Current-Reuse Wideband Receiver With Parallel N-Path Active/Passive Mixers and a Single-MOS Pole-Zero LPF

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Fujian; Mak, Pui-In; Martins, Rui P.

被引用次数：40 2014

A 2.4 GHz ZigBee Receiver Exploiting an RF-to-BB-Current-Reuse Blixer plus Hybrid Filter Topology in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

被引用次数：45 2014

A 0.14-mm2 1.4-mW 59.4-dB-SFDR 2.4-GHz ZigBee/WPAN Receiver Exploiting a Split-LNTA+50% LO Topology in 65-nm CMOS

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

被引用次数：40 2014

A Sine-LO Square-Law Harmonic-Rejection Mixer-Theory, Implementation, and Application

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Lin, Fujian; Mak, Pui-In; Martins, Rui P.

被引用次数：6 2014

Analysis and Modeling of a Gain-Boosted N-Path Switched-Capacitor Bandpass Filter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lin, Zhicheng; Mak, Pui-In; Martins, Rui P.

被引用次数：74 2014

A 0.137 mm2 9 GHz Hybrid Class-B/C QVCO With Output Buffering in 65 nm CMOS

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Amin, Md. Tawfiq; Mak, Pui-In; Martins, Rui P.

被引用次数：5 2014

Enhancing the performances of recycling folded cascode OpAmp in nanoscale CMOS through voltage supply doubling and design for reliability

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Mak, Pui-In; Liu, Miao; Zhao, Yaohua; Martins, Rui P.

被引用次数：4 2014

An intelligent digital microfluidic system with fuzzy-enhanced feedback for multi-droplet manipulation

LAB ON A CHIP

Gao, Jie; Liu, Xianming; Chen, Tianlan; Mak, Pui-In; Du, Yuguang; Vai, Mang-I; Lin, Bingcheng; Martins, Rui P.

被引用次数：61 2013

A 53-to-75-mW, 59.3-dB HRR, TV-Band White-Space Transmitter Using a Low-Frequency Reference LO in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Un, Ka-Fai; Mak, Pui-In; Martins, Rui P.

被引用次数：11 2013

A 0.016 mm2 144-μW Three-Stage Amplifier Capable of Driving 1-to-15 nF Capacitive Load With >0.95-MHz GBW (vol 48, pg 527, 2013)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Zushu; Mak, Pui-In; Law, Man-Kay; Martins, R. P.

被引用次数：0 2013

Construction of a microfluidic chip, using dried-down reagents, for LATE-PCR amplification and detection of single-stranded DNA

LAB ON A CHIP

Jia, Yanwei; Mak, Pui-In; Massey, Conner; Martins, Rui P.; Wangh, Lawrence J.

被引用次数：18 2013

15-nW Biopotential LPFs in 0.35-μm CMOS Using Subthreshold-Source-Follower Biquads With and Without Gain Compensation

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Zhang, Tan-Tan; Mak, Pui-In; Vai, Mang-I; Mak, Peng-Un; Law, Man-Kay; Pun, Sio-Hang; Wan, Feng; Martins, Rui P.

被引用次数：54 2013

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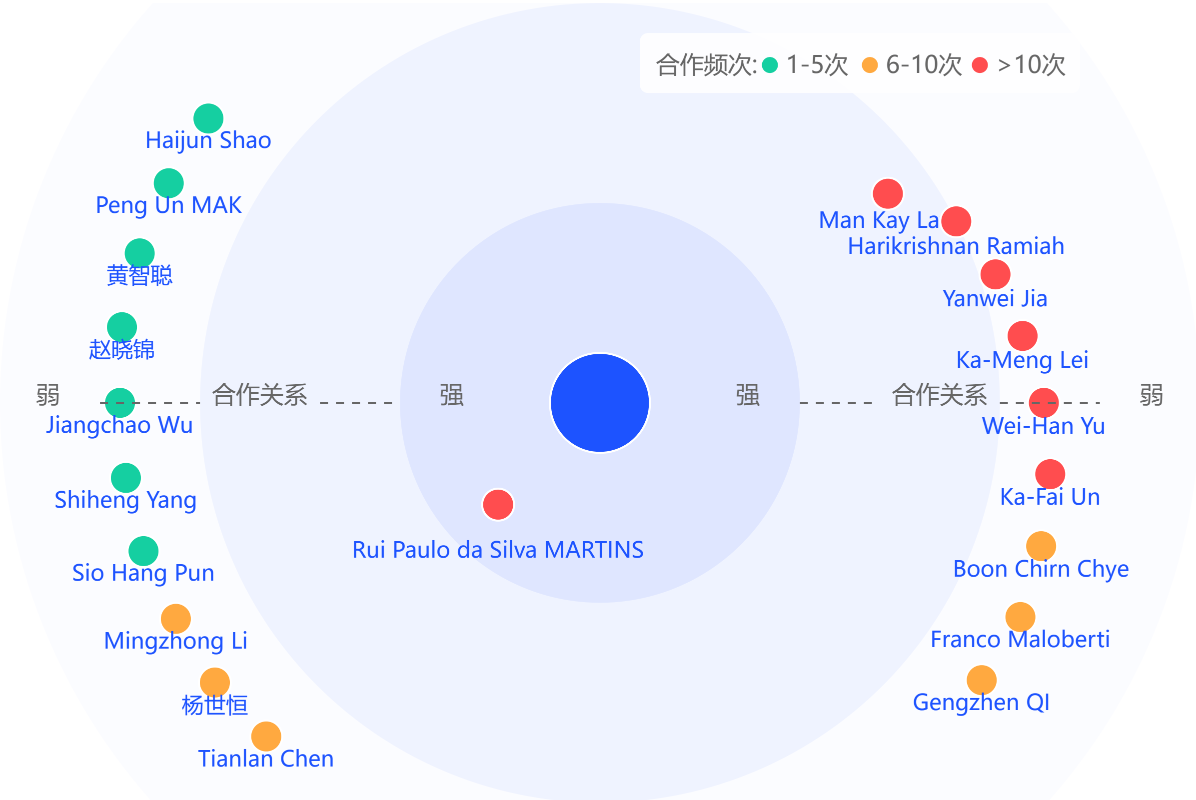

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